

Chapter 1

Spacetime

1.1 Spacetime and notation

We package time and space coordinates into a single four-component object

$$(ct, x, y, z) \equiv x^\mu \quad , \quad x^0, x^1, x^2, x^3 \quad , \quad \mu = 0, 1, 2, 3 \quad (1.1)$$

labeling t, x, y, z respectively. Throughout, we use the index conventions

$$\mu, \nu, \dots = 0, 1, 2, 3 \text{ for spacetime} \quad , \quad i, j, k, \dots = 1, 2, 3 \text{ for space only} \quad (1.2)$$

1.2 Four-velocity and four-momentum

The four-velocity is defined as

$$u^\mu \equiv \frac{dx^\mu}{d\tau} = (c\gamma, \gamma\vec{v}) \quad (1.3)$$

where τ is proper time, and we used time dilation

$$dt = \gamma d\tau \quad , \quad \gamma \equiv \frac{1}{\sqrt{1 - \beta^2}} \quad , \quad \beta \equiv \frac{v}{c} \quad , \quad \vec{v} = \frac{d\vec{x}}{dt} \quad (1.4)$$

Recall the Lorentz transformations relating an inertial frame $\mathcal{O}$ to a frame $\mathcal{O}'$ moving with speed v along the shared x -axis, as depicted in the Figure

$$ct' = \gamma(ct - \beta x) \quad , \quad x' = \gamma(x - \beta ct) \quad , \quad y' = y \quad , \quad z' = z \quad (1.5)$$

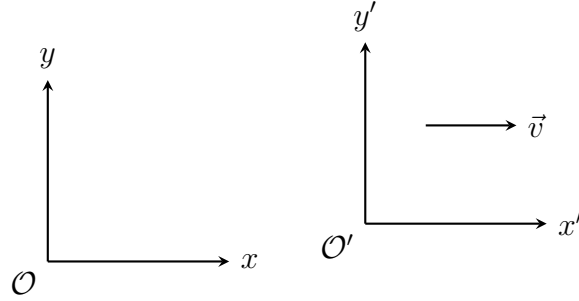


Figure 1.1: Two inertial frames: $\mathcal{O}'$ moves with velocity $\vec{v}$ along the shared x -axis.

which implies for the four-velocity

$$u^{0'} = \gamma(u^0 - \beta u^1) \quad , \quad u^{1'} = \gamma(u^1 - \beta u^0) \quad , \quad u^{2'} = u^2 \quad , \quad u^{3'} = u^3 \quad (1.6)$$

(like a rotation in spacetime). Generally, a 4-vector transforms as

$$u^{\mu'} = \Lambda^{\mu'}_{\mu} u^{\mu} \quad (1.7)$$

using the Einstein summation convention: repeated indices in a given expression are summed over. In matrix form,

$$\begin{pmatrix} u^{0'} \\ u^{1'} \\ u^{2'} \\ u^{3'} \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u^0 \\ u^1 \\ u^2 \\ u^3 \end{pmatrix} \quad (1.8)$$

where we denote the 4×4 matrix $\Lambda^{\mu'}_{\mu} \rightarrow \hat{\Lambda}$, with the upper (primed) index labeling the row and the lower index labeling the column.

The four-momentum is then

$$p^{\mu} \equiv m u^{\mu} = (\gamma m c, \gamma m \vec{v}) = \left(\frac{U}{c}, \gamma m \vec{v} \right) \quad (1.9)$$

where

$$U \equiv \text{energy} = \gamma m c^2 \quad , \quad \vec{p} = \gamma m \vec{v} \quad (1.10)$$

Similarly, the four-force is defined as

$$f^{\mu} \equiv \frac{dp^{\mu}}{d\tau} = \gamma \left(\frac{1}{c} \frac{dU}{dt}, \frac{d\vec{p}}{dt} \right) \quad (1.11)$$

that is

$$f^0 = \frac{\gamma}{c} \frac{dU}{dt} \quad , \quad \vec{f} = \gamma \frac{d\vec{p}}{dt} = \gamma \vec{F} \quad (1.12)$$

with $\vec{F}$ being the usual Newtonian force.

1.3 The metric

Generally, Lorentz transformations are defined by

$$\hat{\Lambda}^T \hat{\eta} \hat{\Lambda} = \hat{\eta} \quad (1.13)$$

where

$$\hat{\eta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (1.14)$$

is called the metric. This condition comes from the postulate that the speed of light is a law of physics and hence is the same in all inertial frames. Along a light ray, we then must have

$$-c^2 dt^2 + dx^2 + dy^2 + dz^2 = 0 = -c^2 dt'^2 + dx'^2 + dy'^2 + dz'^2 \quad (1.15)$$

More generally, for a massive particle, the proper time interval

$$-c^2 d\tau^2 = dx^\mu dx^\nu \eta_{\mu\nu} = dx^{\mu'} dx^{\nu'} \eta_{\mu'\nu'} \quad (1.16)$$

is frame-independent, where $\eta_{\mu'\nu'}$ is the same matrix of numbers as $\eta_{\mu\nu}$. To see how this leads to (1.13), use $dx^{\mu'} = \Lambda^{\mu'}_{\mu} dx^\mu$ to write

$$\begin{aligned} dx^{\mu'} \eta_{\mu'\nu'} dx^{\nu'} &= dx^\mu \Lambda^{\mu'}_{\mu} \eta_{\mu'\nu'} \Lambda^{\nu'}_{\nu} dx^\nu = dx^T \hat{\Lambda}^T \hat{\eta} \hat{\Lambda} dx \\ &= dx^T \hat{\eta} dx = dx^\mu \eta_{\mu\nu} dx^\nu \Rightarrow \hat{\Lambda}^T \hat{\eta} \hat{\Lambda} = \hat{\eta} \end{aligned} \quad (1.17)$$

Hence, Lorentz invariants are constructed by “contraction” with the metric. For example

$$p^\mu \eta_{\mu\nu} p^\nu = p^{\mu'} \eta_{\mu'\nu'} p^{\nu'} = -\left(\frac{U}{c}\right)^2 + \vec{p}^2 = -m^2 c^2 \quad (1.18)$$

1.4 Co-vectors

Define the inverse transformation $\hat{\Lambda}^{-1} \rightarrow \Lambda^\mu_{\mu'}$ – upper index unprimed, lower index primed – taking us from $\mathcal{O}'$ back to $\mathcal{O}$

$$\hat{\Lambda}^{-1}(\beta) = \hat{\Lambda}(-\beta) = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (1.19)$$

We call “co-vector” an object that transforms as

$$b_{\mu'} = \Lambda^\mu_{\mu'} b_\mu \quad (1.20)$$

– note the lower index. Then we can construct Lorentz invariants by

$$a^{\mu'} b_{\mu'} = \Lambda^{\mu'}_{\mu} a^{\mu} \Lambda^{\nu}_{\mu'} b_{\nu} = \delta^{\nu}_{\mu} a^{\mu} b_{\nu} = a^{\mu} b_{\mu} \quad (1.21)$$

where δ^{ν}_{μ} is the identity, with entries 0 or 1.